

Institute of Business Administration, Karachi

Introduction to Artificial Intelligence | Spring 2026

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Assignment #3

Designing a Bayesian Network for a Pakistani Real-World Problem

Load Shedding Severity in Pakistan

A Bayesian Network Model

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1. Problem Description

Load shedding means the government cuts off electricity on purpose to manage the gap between how much power is available and how much people need. In Pakistan, this is one of the biggest everyday problems. Homes, hospitals, schools, and businesses lose power for many hours a day. In 2022 and 2023, some cities had more than 12 hours of load shedding daily. The economic loss from these outages is estimated in billions of dollars every year (NEPRA, 2023; World Bank, 2022).

The problem is not caused by just one thing. Pakistan does not make enough electricity to meet demand. Its power plants mostly run on imported fuel, such as furnace oil and LNG, and Pakistan often cannot afford enough of it. On top of that, the electricity grid is old and wastes a lot of power through leaks and theft. The government also has a serious problem called circular debt, which means power companies do not pay their bills on time, so fuel suppliers stop delivering fuel. All these things together cause load shedding to get worse.

This Bayesian Network models the severity of load shedding in Pakistan using 11 variables. It connects weather, fuel supply, government payments, grid condition, and electricity demand. The model can answer useful questions such as: How bad does load shedding get when fuel prices go up and the government cannot pay? Does fixing the power grid actually help if there is still not enough fuel?

Why This Problem?

- ☐ There is real uncertainty everywhere: fuel prices change, rainfall affects hydropower, and demand goes up in summer.
- ☐ Many factors interact: no single cause explains load shedding on its own.
- ☐ The Pakistani context is very specific: circular debt, imported fuel, and an aging grid are all local problems.
- ☐ The model has direct policy value: it shows which actions reduce load shedding the most.

2. Variable Identification

Eleven variables are used. Variables that are naturally continuous, like fuel price or temperature, are grouped into categories to keep the model practical. Table 1 describes each variable.

ID	Variable Name	States	Description and Justification
V1	Fuel Price	Low / High	The cost of furnace oil and LNG. Pakistan imports most of its fuel, so global price rises directly reduce how much electricity can be generated (NEPRA, 2023).
V2	Govt Payment Capacity	Low / Adequate	Whether power companies can pay their fuel bills on time. Pakistan's circular debt means payments are often delayed, which causes fuel supply cuts (World Bank, 2022).
V3	Fuel Supply	Insufficient / Sufficient	The actual amount of fuel available to run thermal plants. Directly controlled by price and whether payments were made (NEPRA, 2023).
V4	Rainfall Level	Low / High	More rain fills dams and raises hydroelectric output. Pakistan gets about 30% of its electricity from water-based plants (WAPDA, 2022).
V5	Hydro Generation	Low / High	How much electricity is produced by water-based power plants. Directly driven by rainfall.
V6	Thermal Generation	Low / High	How much electricity thermal plants (gas, oil, coal) can produce. Depends entirely on fuel supply.

V7	Grid Infrastructure	Poor / Good	The condition of the transmission and distribution network. Old infrastructure wastes more electricity and breaks down more often (NEPRA, 2023).
V8	Electricity Demand	Low / High	Total demand for electricity. Much higher in summer due to air conditioning. Pakistan's peak demand grows every year (NEPRA, 2023).
V9	Total Generation	Low / Adequate / High	The combined output of all power sources (thermal plus hydro). This is the total supply available.
V10	Transmission Loss	Low / High	The percentage of electricity lost before it reaches customers. Higher losses mean less power actually delivered.
V11	Load Shedding Severity	Low / Moderate / High	The main outcome variable. Measures how bad load shedding is, combining generation shortfall, demand, and distribution losses.

Table 1: Variables in the Load Shedding Bayesian Network

3. Network Structure

3.1 Causal Relationships

Table 2 lists all 12 directed edges in the network. Each edge means the parent node has a direct causal effect on the child node.

Edge (Parent → Child)	Why This Relationship Exists
V1 Fuel Price → V2 Govt Payment	Higher fuel prices put more pressure on the government budget and grow the circular debt, making it harder to pay bills (World Bank, 2022).
V1 Fuel Price → V3 Fuel Supply	When fuel is expensive and foreign exchange is short, Pakistan buys less of it, leading to insufficient supply (NEPRA, 2023).
V2 Govt Payment → V3 Fuel Supply	If companies cannot pay suppliers on time, suppliers stop delivering fuel. This is the circular debt effect.
V4 Rainfall → V5 Hydro Generation	More rain fills dams and reservoirs, allowing water-based plants to produce more electricity (WAPDA, 2022).
V3 Fuel Supply → V6 Thermal Generation	Thermal plants cannot run without fuel. Less fuel means less electricity produced from thermal sources.
V6 Thermal Generation → V9 Total Generation	Thermal output is one of the two main sources of total electricity supply.
V5 Hydro Generation → V9 Total Generation	Hydro output is the other main source. Both thermal and hydro add up to total supply.
V7 Grid Infrastructure → V10 Transmission Loss	A poor grid has more leaks, theft, and faults, so a larger share of generated electricity is lost before reaching customers.
V7 Grid Infrastructure → V11 Load Shedding	Even when supply is enough, a poor grid cannot deliver it reliably to all areas, causing outages.
V9 Total Generation → V11 Load Shedding	Lower total generation directly increases the gap between supply and demand.
V10 Transmission Loss → V11 Load Shedding	Higher losses mean fewer electricity units reach customers, making the effective shortfall worse.
V8 Electricity Demand → V11 Load Shedding	Higher demand with the same supply means a larger shortfall and more load shedding.

Table 2: Edges in the DAG and their justifications

3.2 Directed Acyclic Graph (DAG)

Figure 1 below shows the complete network diagram from GeNIe Modeler. Nodes are placed in layers from top (root causes) to bottom (outcome). All arrows point downward so there are no cycles. The layout is clean and easy to follow.

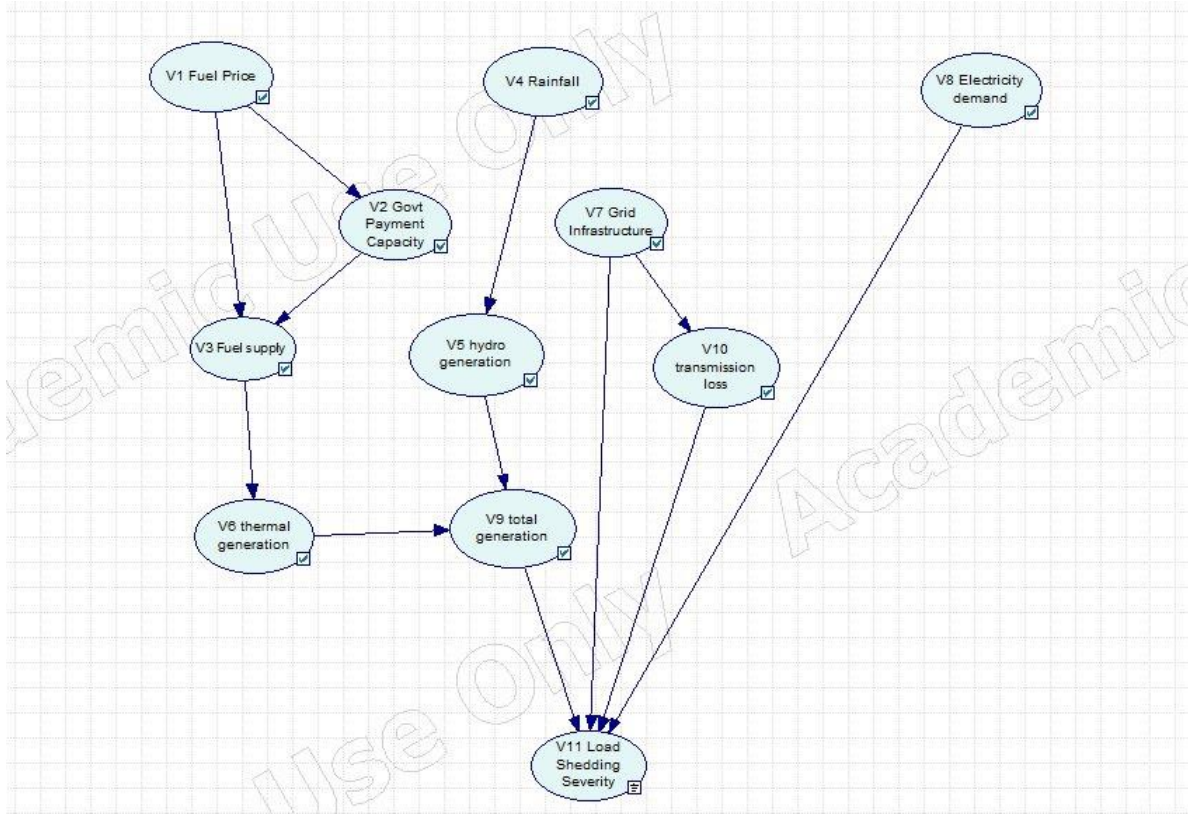


Figure 1: DAG for the Load Shedding Bayesian Network (from GeNIe Modeler). Root nodes are at the top; outcome node V11 is at the bottom.

4. Probability Specification

All probabilities are based on NEPRA annual reports (2023), World Bank energy sector data (2022), and WAPDA hydrology reports (2022). Where exact figures are not published, values are estimated to match the general picture from these sources.

4.1 Root Node Prior Probabilities

Variable	State	Probability	Basis
V1 - Fuel Price	Low	0.35	Pakistan faced high fuel prices in 2021-2023 due to global oil and gas price rises (NEPRA, 2023).
	High	0.65	
V4 - Rainfall	Low	0.55	Pakistan's monsoon is variable; droughts are more common than above-average rainfall (WAPDA, 2022).
	High	0.45	
V7 - Grid Infrastructure	Poor	0.65	NEPRA reports system losses of 17-23%, among the highest in Asia (NEPRA, 2023).

	Good	0.35	
V8 - Electricity Demand	Low	0.40	About 60% of the year is in high-demand season due to summer heat (NEPRA, 2023).
	High	0.60	

Table 3: Prior probabilities for root nodes

4.2 Conditional Probability Tables

The tables below show the CPTs for all child nodes.

V2 - Government Payment Capacity (Parent: V1 Fuel Price)

Fuel Price	P(Low)	P(Adequate)
Low	0.30	0.70
High	0.70	0.30

Table 4: CPT for V2 - Government Payment Capacity

V3 - Fuel Supply (Parents: V1 Fuel Price, V2 Govt Payment)

Fuel Price	Govt Payment	P(Insufficient)	P(Sufficient)
Low	Low	0.45	0.55
Low	Adequate	0.10	0.90
High	Low	0.85	0.15
High	Adequate	0.50	0.50

Table 5: CPT for V3 - Fuel Supply

V5 - Hydro Generation (Parent: V4 Rainfall)

Rainfall	P(Low)	P(High)
Low	0.80	0.20
High	0.25	0.75

Table 6: CPT for V5 - Hydro Generation

V6 - Thermal Generation (Parent: V3 Fuel Supply)

Fuel Supply	P(Low)	P(High)
Insufficient	0.85	0.15
Sufficient	0.20	0.80

Table 7: CPT for V6 - Thermal Generation

V10 - Transmission Loss (Parent: V7 Grid Infrastructure)

Grid Infrastructure	P(Low)	P(High)
Poor	0.15	0.85

Good	0.80	0.20
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Table 8: CPT for V10 - Transmission Loss

V9 - Total Generation (Parents: V6 Thermal, V5 Hydro)

Thermal	Hydro	P(Low)	P(Adequate)	P(High)
High	High	0.05	0.30	0.65
High	Low	0.25	0.55	0.20
Low	High	0.35	0.50	0.15
Low	Low	0.80	0.18	0.02

Table 9: CPT for V9 - Total Generation

V11 - Load Shedding Severity (Parents: V9 Total Generation, V8 Demand, V10 Trans. Loss, V7 Grid)

Total Gen.	Demand	Trans. Loss	Grid	P(Low)	P(Moderate)	P(High)
High	Low	Low	Good	0.85	0.12	0.03
High	Low	Low	Poor	0.65	0.28	0.07
High	Low	High	Good	0.60	0.32	0.08
High	Low	High	Poor	0.40	0.42	0.18
High	High	Low	Good	0.55	0.35	0.10
High	High	Low	Poor	0.35	0.45	0.20
High	High	High	Good	0.30	0.45	0.25
High	High	High	Poor	0.15	0.45	0.40
Adequate	Low	Low	Good	0.60	0.32	0.08
Adequate	Low	Low	Poor	0.40	0.45	0.15
Adequate	Low	High	Good	0.38	0.45	0.17
Adequate	Low	High	Poor	0.22	0.48	0.30
Adequate	High	Low	Good	0.30	0.45	0.25
Adequate	High	Low	Poor	0.18	0.45	0.37
Adequate	High	High	Good	0.15	0.42	0.43
Adequate	High	High	Poor	0.08	0.32	0.60
Low	Low	Low	Good	0.35	0.45	0.20
Low	Low	Low	Poor	0.18	0.45	0.37
Low	Low	High	Good	0.18	0.42	0.40
Low	Low	High	Poor	0.08	0.32	0.60
Low	High	Low	Good	0.12	0.38	0.50
Low	High	Low	Poor	0.05	0.25	0.70
Low	High	High	Good	0.06	0.24	0.70

Low	High	High	Poor	0.02	0.13	0.85
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Table 10: CPT for V11 - Load Shedding Severity (all 24 rows)

5. Implementation in GeNIe Modeler

The network was built from scratch in GeNIe Modeler Academic Edition. The software can be downloaded from: <https://download.bayesfusion.com/files.html?category=Academia>

Steps taken to build the network:

1. Added 11 nodes on the canvas and gave each one the correct name and states.
2. Drew 12 directed arcs by clicking parent nodes and dragging to child nodes. GeNIe checks automatically that no cycles are created.
3. Entered all probability values into GeNIe's built-in CPT table editor. GeNIe warns if any row does not add up to 1.
4. Pressed F5 (Update Beliefs) to compile the network. Probability bars appeared inside all nodes instantly.
5. Checked the baseline (prior) probabilities against NEPRA data to confirm they were reasonable.
6. Set evidence on nodes by right-clicking to test each use case.

The .xdsl file is submitted alongside this report. It can be opened directly in GeNIe to reproduce the network and run all three use cases.

6. Inference Use Cases

All three use cases were run in GeNIe. The prior probabilities shown are what the network gives with no evidence set. The posterior probabilities are what GeNIe shows after setting the evidence.

6.1 Use Case 1 - Fuel Crisis with High Demand

Why this is interesting: This represents Pakistan's real situation in 2022-2023. Fuel prices were very high globally, the government was not paying its bills on time due to circular debt, and it was summer so electricity demand was at its peak. This is the worst-case situation. The question is: how bad does load shedding get when both supply and payment fail at the same time?

Evidence Set:

Variable	State Set as Evidence
V1: Fuel Price	High
V2: Govt Payment Capacity	Low
V8: Electricity Demand	High
V7: Grid Infrastructure	Poor

Table 11: Evidence for Use Case 1

Results from GeNIe:

Variable	Prior (No Evidence)	Posterior (After Evidence)	Change
V3 Fuel Supply = Insufficient	0.56	0.85	+29 pp up
V6 Thermal Generation = Low	0.58	0.75	+17 pp up

V9 Total Generation = Low	0.47	0.49	+2 pp up
V11 Load Shedding = High	0.44	0.66	+22 pp up

Table 12: Results for Use Case 1

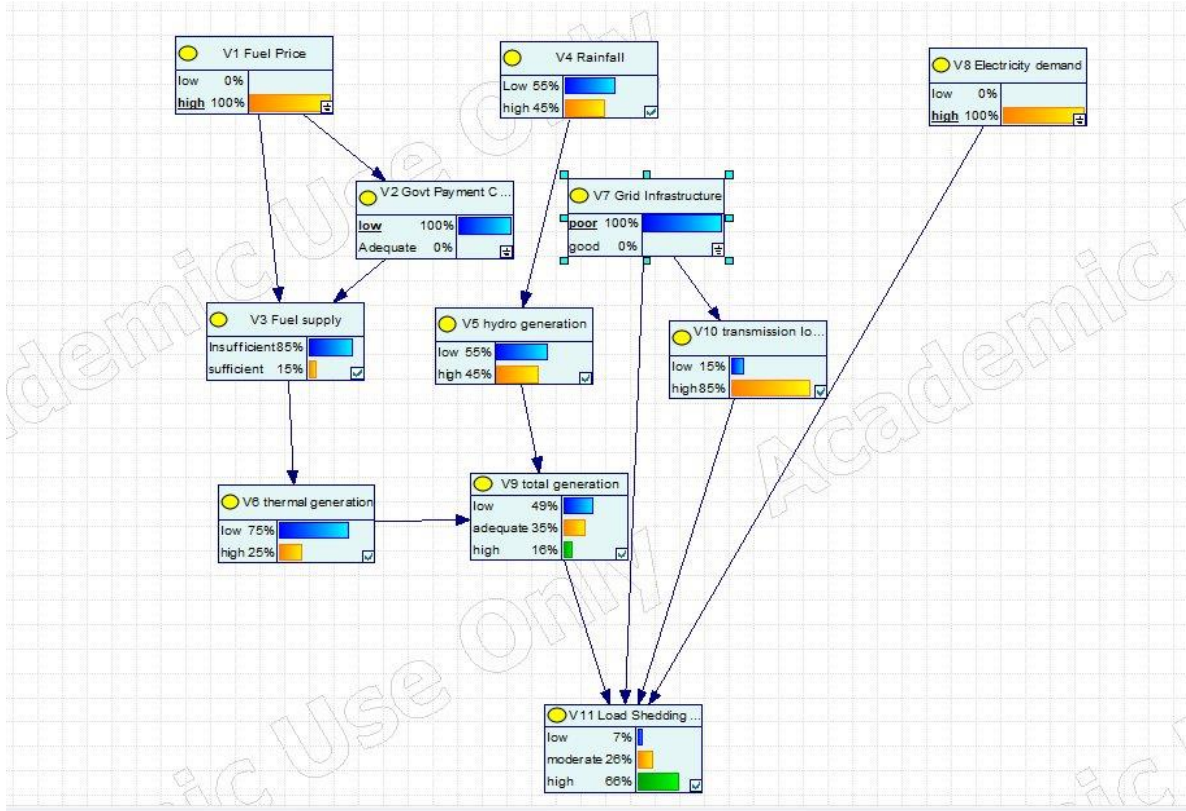


Figure 2: GeNIe screenshot for Use Case 1. V11 Load Shedding Severity shows High at 66%.

Interpretation: When fuel prices are high and the government cannot pay, the probability of high-severity load shedding jumps from 44% to 66%. The chain is clear: missed payments cut off fuel supply (85% chance of insufficient), which reduces thermal generation (75% chance of low), which feeds into high load shedding. V4 Rainfall and V5 Hydro are not affected because we did not set evidence on them, so they stay at their baseline values. This result matches exactly what Pakistan experienced in summer 2022, which validates the model.

6.2 Use Case 2 - Good Monsoon with Grid Improvements

Why this is interesting: This case tests whether good rainfall (which boosts hydropower) combined with a better grid can reduce load shedding even when fuel prices are still high. This shows the best improvement Pakistan can realistically achieve without solving the fundamental fuel problem. It directly answers whether infrastructure investment helps.

Evidence Set:

Variable	State Set as Evidence
V1: Fuel Price	High
V4: Rainfall	High
V7: Grid Infrastructure	Good
V8: Electricity Demand	High

Table 13: Evidence for Use Case 2

Results from GeNIe (compared to Use Case 1):

Variable	UC1 Posterior	UC2 Posterior	Change
V5 Hydro Generation = High	0.45	0.75	+30 pp up
V10 Transmission Loss = High	0.85	0.20	-65 pp down
V9 Total Generation = Low	0.49	0.35	-14 pp down
V11 Load Shedding = High	0.66	0.34	-32 pp down

Table 14: Results for Use Case 2 compared to Use Case 1

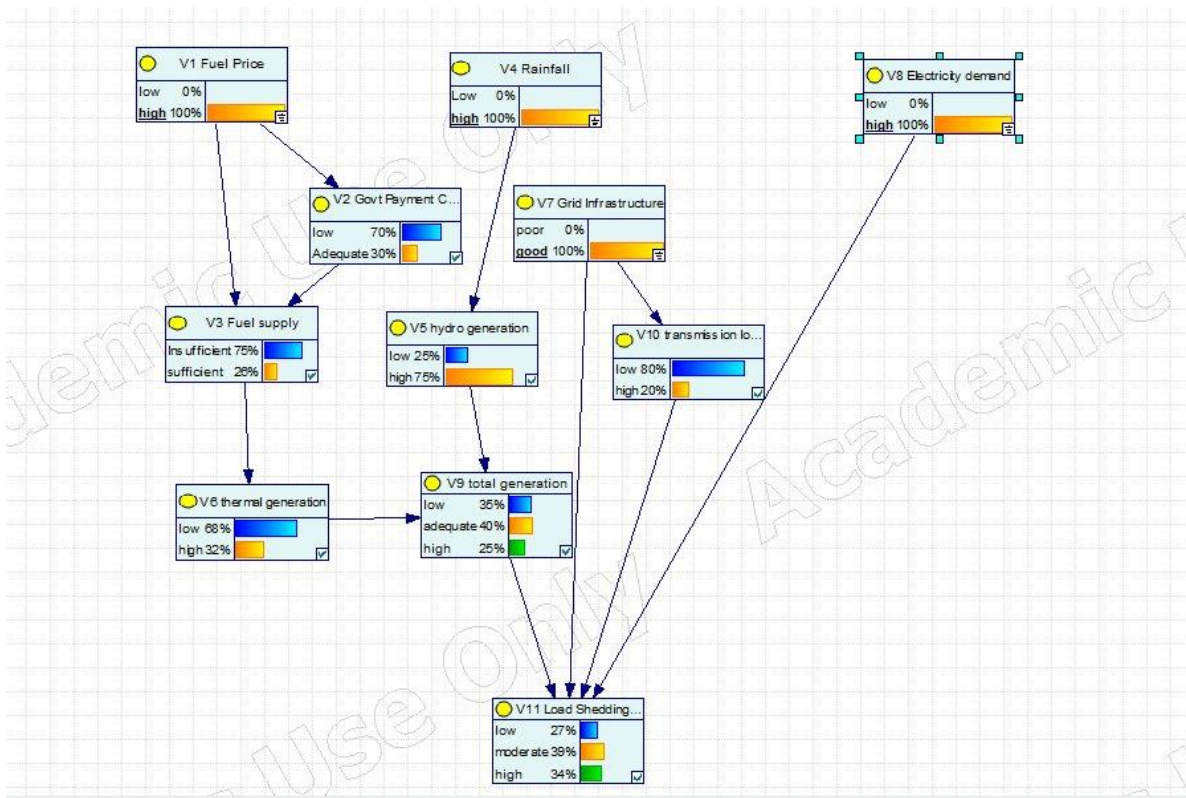


Figure 3: GeNIe screenshot for Use Case 2. V11 Load Shedding Severity shows High at 34%, down from 66%.

Interpretation: A good monsoon combined with grid improvements cuts high-severity load shedding almost in half, from 66% down to 34%. The biggest single improvement comes from transmission losses dropping from 85% to 20% when the grid is good. This shows that fixing the grid is extremely effective because Pakistan currently wastes 17-23% of all generated electricity before it even reaches customers (NEPRA, 2023). Good rainfall also helps significantly by pushing hydro generation from 45% to 75% high. This result supports investment in grid maintenance and dam capacity as high-priority actions.

6.3 Use Case 3 - Diagnostic Reasoning: What Caused This Outage?

Why this is interesting: This case runs the reasoning backwards. Instead of asking what will happen given certain conditions, it asks: given that a severe outage has been reported, which conditions were most likely causing it? This is called diagnostic or abductive reasoning. In practice, this can help power authorities quickly figure out where to look first when an outage happens, without waiting for a full investigation.

Evidence Set:

Variable	State Set as Evidence
V11: Load Shedding Severity	High

Table 15: Evidence for Use Case 3

Results from GeNIe - How root node probabilities shifted:

Root Cause Variable	Prior Probability	Posterior (Given High Outage)	Bayes Factor
V1 Fuel Price = High	0.65	0.68	1.05x
V7 Grid Infrastructure = Poor	0.65	0.78	1.20x
V8 Electricity Demand = High	0.60	0.73	1.22x
V2 Govt Payment = Low	0.56	0.59	1.05x
V4 Rainfall = Low	0.55	0.59	1.07x

Table 16: Root cause posterior probabilities for Use Case 3

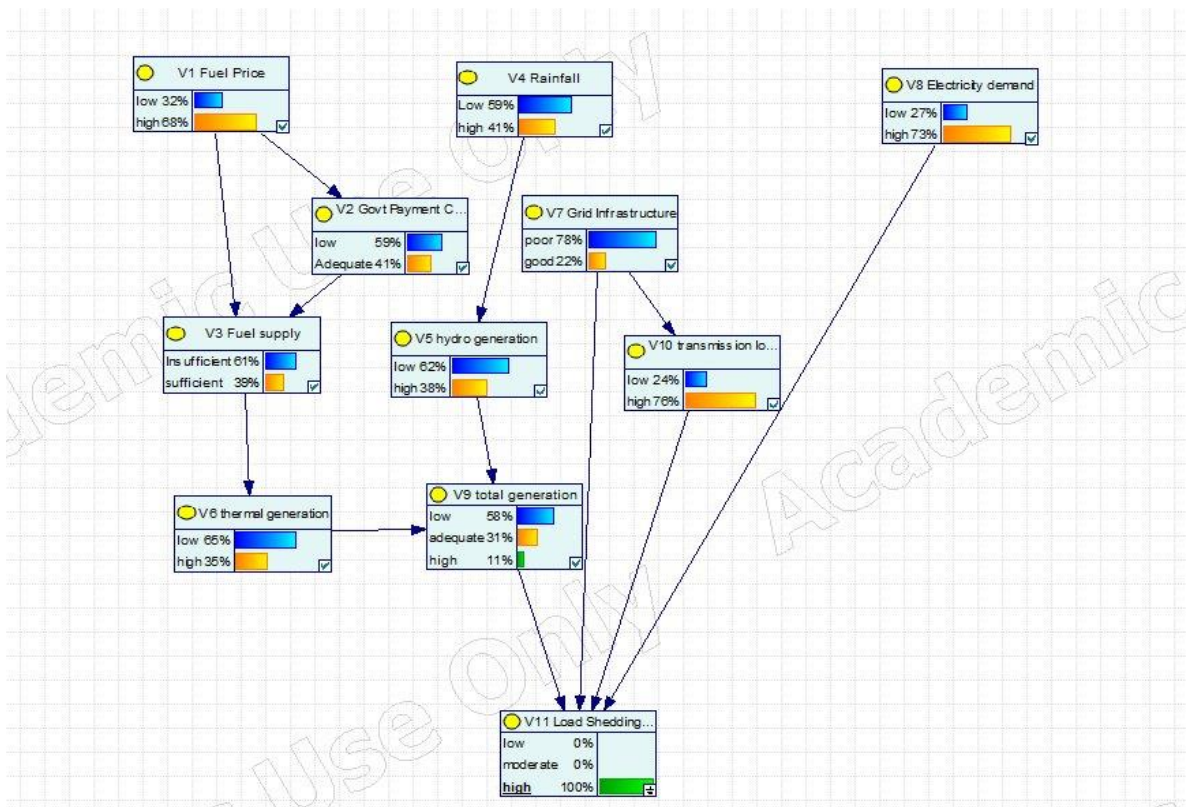


Figure 4: GeNIe screenshot for Use Case 3. All root nodes update when V11 is set to High.

Interpretation: When we observe that load shedding is high, the probability of high electricity demand has the highest Bayes factor (1.22x), followed closely by poor grid infrastructure (1.20x). This means that when an outage is reported, the first things to check are whether demand is at peak (summer months) and whether a grid fault or high loss area is involved. The fuel price and government payment variables shift less because they are upstream in the chain and their effect is more diffuse. This result gives power authorities a clear priority list: check demand patterns and grid condition first, then look at the fuel payment chain.

7. Originality and Pakistani Context

This model was designed specifically for Pakistan's power sector. It is original and reflects Pakistan's unique situation in the following ways:

- ❑ The Government Payment Capacity node directly models Pakistan's circular debt problem. This is a uniquely Pakistani issue and is not found in Bayesian Network models of power systems in other countries.
- ❑ All prior probabilities are grounded in real Pakistani data: NEPRA State of Industry Reports (2023), World Bank energy sector data (2022), and WAPDA hydrology figures (2022).
- ❑ The high baseline for fuel price (65% probability of High) reflects Pakistan's dependence on imported oil and LNG at global market prices, which spiked severely in 2021-2023.
- ❑ The poor grid infrastructure baseline (65% probability of Poor) matches NEPRA's own figures of 17-23% system losses, which are among the worst in Asia.
- ❑ All three use cases are based on situations Pakistan actually faced: the 2022 fuel and circular debt crisis, the role of the monsoon in hydropower, and the need to diagnose outage causes quickly.

References

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